Twinkl: An inner compass that helps users align behavior with long-term values

Capstone 5 - Project Proposal (Apr 2026)

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1. Introduction

Problem Statement

Ambitious people often know what matters to them, but their weekly behaviour drifts quietly away from those priorities. Health, relationships, meaningful work, and personal direction are easy to state and hard to sustain under ordinary pressure. Existing journaling products help users log feelings, answer prompts, and review trends, but they rarely maintain an explicit model of what the user said they care about and then evaluate later behaviour against it.

Twinkl addresses that gap. The product treats declared priorities as system state, then compares new journal behaviour against that state through the Value Identity Function (VIF): a multi-dimensional, uncertainty-aware evaluator built around Schwartz's ten-value framework. The intended result is an evidence-grounded weekly reflection on whether the user's actions still match the direction they claimed to want, rather than a better mood summary.

For this milestone, the core question is whether this alignment engine can be built, trained, validated, and run in a way that justifies continued investment.

Milestone Status

This milestone already answers four practical questions:

1. The offline teacher-student pipeline can be built at usable scale.
2. Human calibration provides a benchmark for judge supervision at POC scale.
3. A compact critic can learn meaningful alignment signal from the available data.
4. The project already has an internal runtime path from model checkpoint to weekly output.

The main remaining risk is whether the value model is reliable enough to support trustworthy user-facing feedback beyond synthetic and internal review settings.

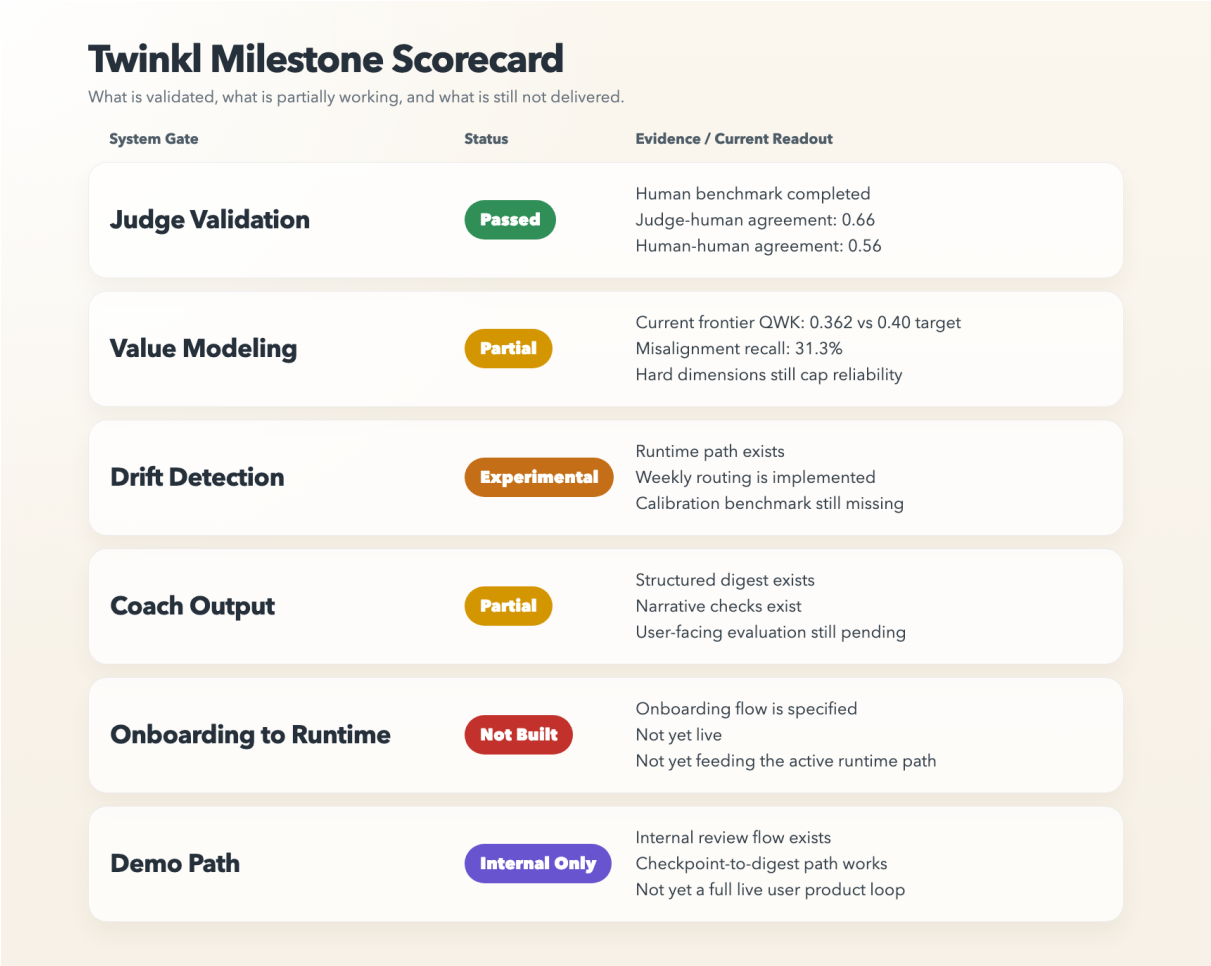


Figure 1: Milestone scorecard

Current delivery state

Delivery state	What it means in this milestone
Implemented now	Synthetic persona generation, judge labeling, annotation workflow, critic training, experiment archive, and an internal checkpoint-to-digest review flow are all working today.
Implemented experimentally / not yet validated	Runtime inference, weekly aggregation, crash/rut routing, weekly digest generation, and first-pass narrative validation can run, but they are not yet calibrated or ready for external demonstration.
Specified but not built	Live onboarding, onboarding-to-runtime integration, real-user journaling orchestration, calibrated weekly detection benchmarks, and a true end-user product loop are not yet delivered.

At this stage, Twinkl remains an internal prototype rather than a user-ready product.

Differentiation and Target Users

Twinkl is aimed first at knowledge workers in transition: graduate students, new managers, founders, and other high-agency professionals managing recurring trade-offs across work, health, relationships, and identity. These users already journal, seek coaching, or pay for reflective productivity tools, yet most available products still stop at summarisation, prompting, or general wellness encouragement.

The product distinction is straightforward: most AI journaling tools explain what the user has been feeling; Twinkl is designed to ask whether the user has been living in line with what they said matters. Several funded competitors illustrate the gap:

App	Core Approach	Gap vs Twinkl
Rosebud	CBT/ACT-based therapeutic reflection with habit tracking	No declared-value model; analyses mental patterns, not behavioural drift from stated priorities
Reflection	AI companion with mood tracking and retrospective theme discovery	Values discovered post-hoc from entries, not proactively declared or tracked over time
Mindsera	50+ generic mental frameworks (CBT, Stoic, Ikigai, etc.) with multi-perspective AI	External frameworks applied uniformly, not adapted to user-specific value trade-offs
Reflectly	Mood tracking with conversational AI and daily challenges	Purely emotional pattern analysis; no values framework or alignment assessment
Life Note	1,000+ AI mentors modelled on historical figures	Mentor philosophies are fixed; no mechanism to track the user's own evolving priorities
Stoic	Stoic philosophy daily rituals with GPT-4	Single prescriptive philosophy, not a user-declared values model

The addressable market spans digital journal apps (USD 5.69 billion in 2025, 11.5% CAGR), mental health apps (USD 7.48 billion, 14.6% CAGR), and personal development tools (USD 46.1 billion, 8.0% CAGR). Twinkl sits at the intersection: structured value-alignment coaching delivered through journaling.

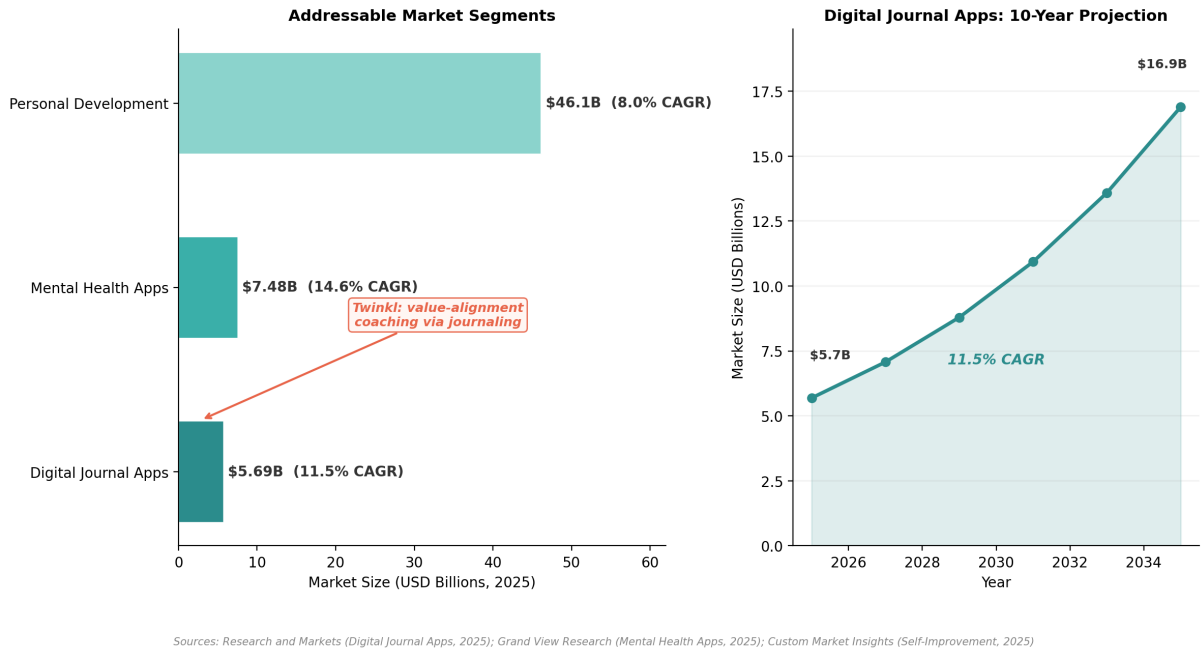


Figure 2: Market context

Related Academic Work

Schwartz Theory of Basic Human Values

Twinkl uses Schwartz's ten-value framework because it provides a practical and well-studied structure for modelling competing priorities rather than treating alignment as a single scalar score (Schwartz, 1992; Schwartz et al., 2012). In this project, the framework is not included as background theory alone; it is the operational schema for persona generation, judge rubrics, and model targets.

Best-Worst Scaling for Value Elicitation

Best-Worst Scaling was chosen for onboarding because it produces clearer trade-offs than Likert-style self-rating and reduces the "everything matters" problem common in value elicitation (Louviere et al., 2015). That matters for Twinkl because the critic is intended to interpret journal behaviour in light of the user's declared priorities, not in light of generic values alone.

Synthetic Data & LLM-as-Judge

Twinkl uses a generate-annotate-learn pipeline because there is no public dataset that pairs longitudinal journal behaviour with per-dimension value alignment labels at the level of detail required here. Synthetic generation provides controlled trajectories, while rubric-guided LLM judging provides scalable supervision that can then be distilled into a smaller runtime model (He et al., 2022; Zheng et al., 2023). Because this supervision is subjective, the project validates it explicitly against human annotation rather than treating synthetic labels as self-justifying.

2. System Architecture

End-to-end Pipeline

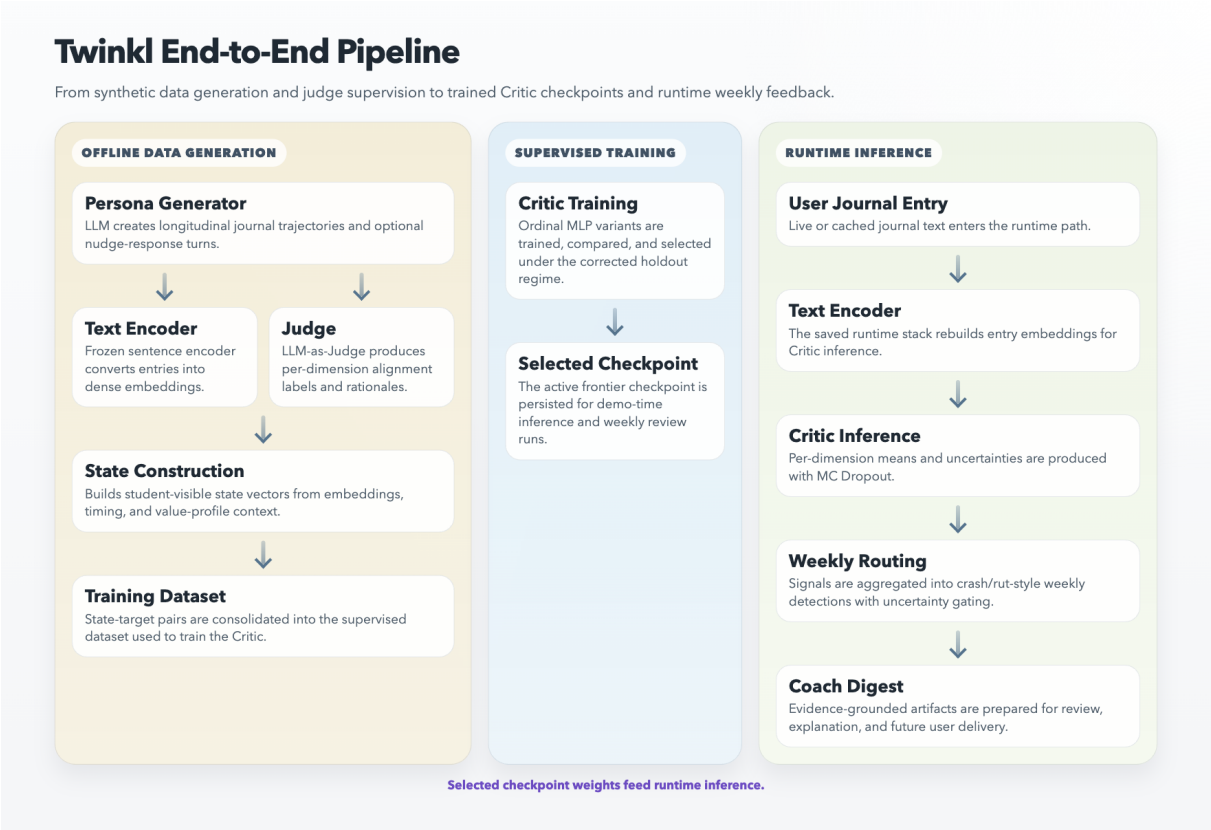


Figure 3: End-to-end pipeline

Twinkl has two linked but distinct system layers. The first is an offline teacher-student pipeline that creates data, labels it, and trains the critic. The second is an experimental runtime path that turns a saved critic checkpoint into weekly alignment signals and coach-ready artifacts. The offline core is already real, while the runtime path is usable internally but not yet fully validated as a product experience.

System layer	Role	Current state
Offline teacher-student pipeline	Generate journal trajectories, label them, and train the critic	Implemented
Runtime scoring and weekly review	Rebuild state, score entries, aggregate weekly signals, route drift modes, and generate digest artifacts	Implemented experimentally
Onboarding and live product loop	Collect user priorities, feed them into runtime state, and support live journaling behaviour	Specified only

At a high level, the online path is organised into three layers:

1. **Critic**: produces per-dimension alignment scores and uncertainty from a compact state representation.
2. **Weekly routing**: aggregates scores into weekly signals and applies uncertainty-gated crash/rut-style logic.
3. **Coach**: converts those signals into evidence-grounded weekly reflections.

Each layer operates over a different scope of information: entry-level state for scoring, weekly aggregation for routing, and broader journal history for explanation.

3. Offline Data Generation and Model Training

Twinkl's offline core is now implemented. The project has a complete teacher-student pipeline covering data generation, labeling, human validation, critic training, and experiment review. As of this milestone, that pipeline operates over **204 synthetic personas** and **1,651 judged journal entries**.

Synthetic Persona Generation

The project requires paired training data: journal entries and per-dimension value-alignment labels. Because no public dataset exists for this problem, the team built a synthetic generation pipeline that produces longitudinal journal trajectories rather than isolated text snippets.

Each persona combines a demographic profile with 1-2 emphasized Schwartz dimensions. Generation runs in parallel across personas but sequentially within each persona so later entries can refer to earlier events. Entries vary in tone and reflection style, and the pipeline includes a conversational nudge system that uses a three-category decision engine (clarification, elaboration, tension-surfacing) with anti-annoyance throttling (maximum two nudges per three entries). Nudge-response pairs are integrated across 62% of the current corpus and better reflect how users interact with modern journaling products.

The pipeline also includes leakage controls. Explicit Schwartz terminology is banned from generated text, generation context is separated from labeling context, and targeted augmentation for weak dimensions is accepted only after judged outcome checks rather than prompt intent alone.

Targeted generation has been especially important for rare and difficult dimensions. Instead of relying only on generic "misaligned" scenarios, the pipeline can produce value-specific tension cases designed to probe the critic's weakest semantic boundaries.

LLM-as-Judge Labeling

The judge labels each journal entry across all ten Schwartz dimensions on a three-way ordinal scale:

Label	Meaning
-1	Misaligned with the value
0	Neutral or no relevant evidence
+1	Aligned with the value

The judge reads the persona biography, prior journal history, and the current session content. This broader context lets it interpret vague entries as part of a trajectory rather than as isolated one-off statements. Each label is accompanied by a short rationale to support later review.

The labeling path is auditable: raw markdown is wrangled into structured entries, labels are produced with schema validation, and the outputs are then consolidated into a deterministic training dataset. A consensus re-judging path also exists as a diagnostic branch so the team can test label stability across multiple independent judge passes.

Human Annotation and Validation

Because the supervision problem is subjective, Twinkl validates judge labels against human scoring rather than assuming that synthetic labels are reliable by construction. The team built a custom annotation application that presents entries sequentially within a persona context and collects per-dimension {-1, 0, +1} judgments.

For the current benchmark, three annotators labeled a shared **115-entry** subset spanning **19 personas**, producing roughly **380 total annotations**. This gives the project a human reference point for judging whether the LLM supervision is usable at POC scale.

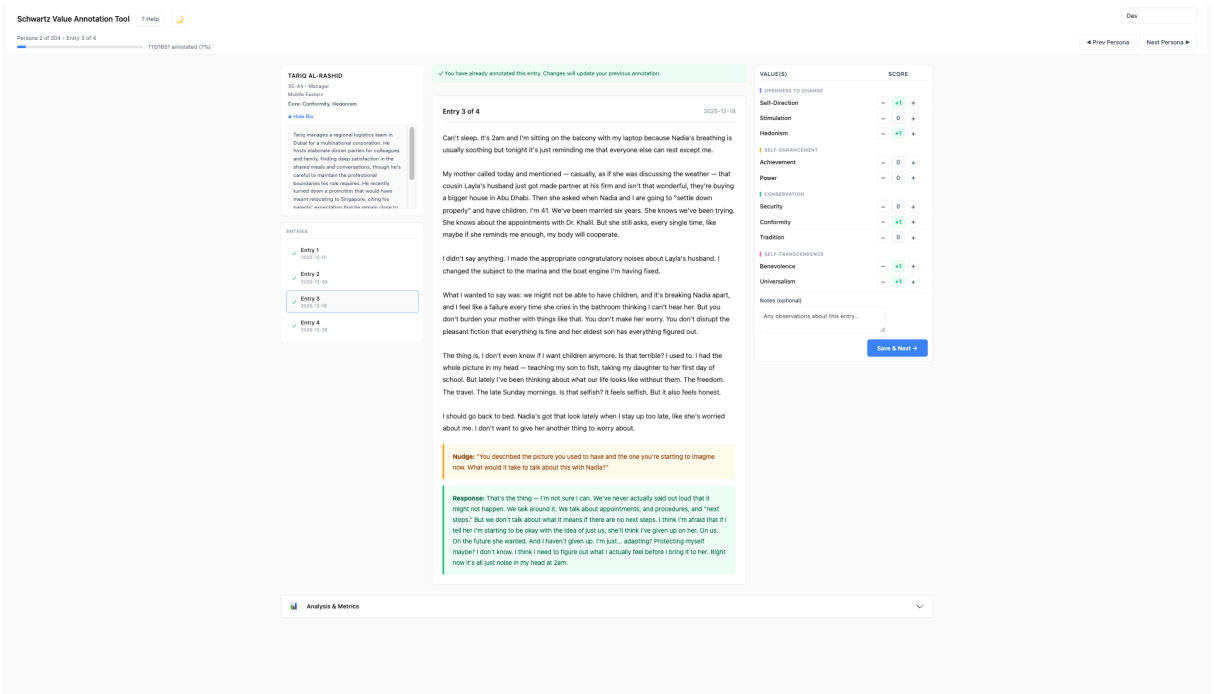


Figure 4: Annotation tool

Critic (VIF) Model Training

The critic is a compact multi-layer perceptron that learns to approximate the judge's per-dimension alignment decisions while remaining cheap enough for runtime inference. This is the core distillation step that turns an expensive, context-heavy teacher into a practical student model.

The current default setup uses:

- a frozen sentence encoder based on Nomic Embed Text v1.5
- Matryoshka truncation to **256 dimensions**
- a compact state representation with **window size = 1**
- a **10-dimensional value profile vector**
- ordinal decoding into {-1, 0, +1}
- **MC Dropout** for uncertainty estimation

The current state intentionally favors reliability over architectural ambition. Larger history windows were explored but overfit badly at current data scale, so the active baseline keeps the runtime state compact and treats short-horizon context as a targeted next-step question rather than a default.

A Bayesian baseline is also implemented as a secondary uncertainty reference. The broader takeaway is unchanged: the project already has a working student model, and the open question is how far that model can be pushed before target quality becomes the real bottleneck.

Training and Experiment Infrastructure

The training workflow is disciplined rather than ad hoc. It includes:

- persona-level splitting to avoid leakage across correlated histories
- fixed holdout manifests for comparable follow-up experiments
- guardrailed checkpoint selection based on ranking quality and minority-signal recovery
- logging of configurations, metrics, and run commentary
- post-hoc threshold tuning on validation data only

These choices reduce the risk of over-reading isolated best-seed results and make follow-up experiments more comparable. The project now has a repeatable evaluation regime, an archived experiment history, and explicit promotion criteria for new challengers.

The codebase maintains **53 test modules** covering all nine ordinal loss functions (CORAL, CORN, EMD, SoftOrdinal, CDW-CE, BalancedSoftmax, Two-Stage BalancedSoftmax, LDAM-DRW, and SLACE), gradient flow and numerical stability, evaluation metrics, coach runtime, nudge decision logic, data wrangling, the demo workbench, and local integration smoke tests. That level of automated coverage supports continued iteration without turning every change into a manual regression risk.

Evaluation Framework

The evaluation suite tracks both aggregate and tail-sensitive behaviour:

Metric	Purpose
QWK (Quadratic Weighted Kappa)	Primary ranking metric for ordinal agreement
Recall_-1	Recovery rate for misalignment labels
MinR (Minority Recall)	Sensitivity to the non-neutral classes overall
Hedging rate	How often the model collapses to neutral predictions
Calibration	Whether reported confidence matches actual correctness

To avoid over-interpreting noise, configurations are compared across three training seeds and reviewed at the family level rather than on single runs. Persona-cluster bootstrap intervals are then used to test whether apparent gains survive holdout uncertainty.

4. Online Model Inference

BWS Onboarding

Twinkl addresses cold-start with a Best-Worst Scaling (BWS) onboarding flow. Six 4-item sets — adapted from PVQ-21 portrait items (Schwartz et al., 2001) as concise first-person mobile cards — produce an ordinal ranking of the 10 Schwartz value dimensions. The top two are designated **core values** (monitored for decline); the remaining eight are **peripheral** (monitored for emergence). A mid-flow mirror presents the emerging ranking for optional refinement, and a goal-selection step (work-life balance, life transition, relationships, health/wellbeing, direction, meaningful work) anchors the Coach's initial framing.

For the Critic, core/peripheral designation is converted into a binary weight vector (equal weight on core, zero on peripheral) concatenated with the text embedding in the State Encoder, conditioning alignment predictions so that ambiguous entries are interpreted against declared priorities.

The BWS flow is fully specified but not yet deployed. A planned extension derives a continuous 10-dimensional weight vector from raw BWS scores, giving every dimension a signal proportional to its rank.

Journal Entry Alignment Classification

Conversational Nudges

Within a session, nudges operate on observable text signals: **clarification** (ambiguous entries), **elaboration** (terse entries), and **tension surfacing** (Critic-detected misalignment on a core value). An anti-annoyance gate suppresses nudges after two have fired in the last three entries.

Per-Dimension Scoring

For each journal entry, the Critic outputs per-dimension alignment scores in $\{-1, 0, +1\}$:

Value	Score	Interpretation	Example
Benevolence	+1	Lived this value	"I dropped everything to help my neighbour move"
Hedonism	0	Not relevant to entry	"I worked extra hours today"
Tradition	−1	Acted against this value	"I snapped at my parents over something they expected me to do"

Each dimension uses an **ordinal classification head** enforcing $-1 < 0 < +1$, preventing the model from assigning high probability to both extremes while giving low probability to the neutral class.

Balanced Softmax loss corrects class priors in the softmax denominator, addressing severe imbalance (75.9% neutral, 7.1% misaligned) without resampling. This shifts the decision boundary so rare misalignment signals surface at inference without inflating false positives — critical because a Critic that suppresses −1 predictions would blind downstream drift detection.

MC Dropout (50 forward passes) estimates epistemic uncertainty, suppressing critiques when confidence is low and triggering clarifying questions instead.

Drift & Evolution

Drift is the gap between declared values and the temporal trend of Critic-observed behaviour, sustained beyond noise thresholds — not a single bad entry but a pattern persisting across weeks and surviving uncertainty gating.

The drift layer operates on **weekly aggregates** of raw per-entry Critic scores, smoothing within-week noise while preserving trends.

Core Value Signals (decline)

Signal	Pattern
Crash	Weekly profile-weighted mean drops >0.5 from previous week
Rut	Alignment stays below −0.4 for ≥3 consecutive low-uncertainty weeks

Peripheral Value Signals (emergence)

Signal	Pattern
Rise	Sustained 0→+1 shift — a previously peripheral value consistently appears, suggesting genuine value evolution

Detection Pipeline

The detector makes a single weekly decision, evaluating triggers in priority order (first match wins):

1. **High Uncertainty Gate** — If overall model uncertainty exceeds 0.3, return immediately. The Coach asks for clarification rather than judging on unreliable scores.
2. **Evolution Classification** — Over a 6-week lookback, compute per-dimension residual (observed minus expected alignment) and volatility (s.d. of scores). Residual < 0.4 → stable; residual ≥ 0.4 with volatility < 0.5 → evolution (genuine shift); residual ≥ 0.4 with volatility ≥ 0.5 → volatile (divergent but unsettled). Evolution dimensions are excluded from crash/rut checks and routed to Coach messaging inviting profile updates.
3. **Rut Detection** — For each non-evolution core dimension (weight ≥ 0.15), flag as rut if all of the last 3 weeks show low alignment.
4. **Crash Detection** — If this week's profile-weighted mean drops >0.5 from last week, flag each non-evolution core dimension where alignment decreased.
5. **Stable** — No trigger fired.

The dual-trigger design was chosen for interpretability: crash maps to "something shifted this week," rut maps to "this has been weighing on you for a while."

Experimental Detector Ensemble

To evaluate whether the dual-trigger approach has blind spots, complementary rule-based detectors were benchmarked:

Detector	Mechanism	Catches
EMA	Running average of misalignment (recent-weighted, negative scores only); alerts when average exceeds threshold	Slow fades (+1→0) that never reach −1
CUSUM	Cumulative sum with slack parameter (k=0.3) absorbing minor noise; jar grows on persistent misalignment, resets to zero on recovery	Intermittent dips (−1,+1,−1,+1) that individually recover but accumulate
Cosine	Cosine similarity between profile weight vector and alignment vector; alerts when vectors oppose	Multi-value reversals where several dimensions shift moderately negative
Control Chart	Per-dimension baseline from early entries; lower control limit = mean − 2 s.d.	Scores nominally neutral but abnormally low for that user (e.g., habitual +1 dropping to 0)
KL Divergence	KL divergence between baseline and rolling 3-entry score distributions (Laplace-smoothed)	Distributional shape changes (e.g., consistent +1 shifting to a 0/−1 mix)

These detectors run over the full synthetic corpus and are visualised in a comparison dashboard. In the absence of human-labelled crisis weeks, cross-detector consensus (≥4 of 6 flagging the same timestep) serves as pseudo-ground-truth.

A direction under exploration is replacing the dual-trigger with consensus-based firing (multiple detectors must agree), reducing blind spots at the cost of interpretability. Proper evaluation requires synthetic crisis-injection timelines with ground-truth labels, which is planned but not yet completed.

Explainable Feedback via Coach

The coach layer converts critic and weekly-routing signals into a digest that is clear, evidence-grounded, and non-judgmental. In the current POC, this takes the form of a weekly artifact that:

- selects focus tensions and strengths
- cites representative journal evidence
- builds a full-context weekly prompt
- can optionally generate a narrative response
- runs first-pass checks on groundedness, non-circularity, and length

The coach selects from four active behavioural modes — **high uncertainty**, **rut**, **mixed state**, and **background strain** — plus a **stable** fallback when no tension is detected. Each mode shapes the tone, evidence selection, and question framing of the weekly digest so the feedback matches the user's current behavioural pattern rather than applying a generic template.

This is further along than a design sketch, but it is not a finished product experience. Structured digest generation is implemented, narrative-generation hooks exist, and internal validation checks exist, but user-facing explanation quality has not yet been proven through real evaluation.

5. Project Evaluation Metrics

Four-Stage Gate Structure

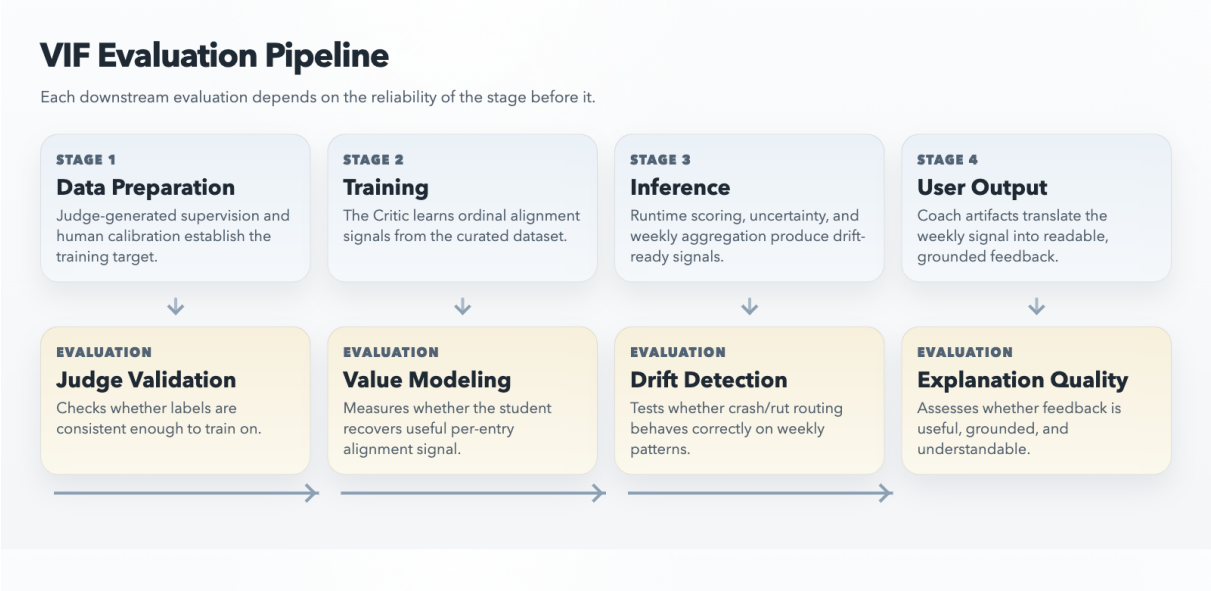


Figure 5: VIF evaluation pipeline

The project uses a sequential evaluation model because every downstream layer depends on the reliability of the layer before it. User-facing weekly feedback is only as trustworthy as the supervision, critic modelling, and weekly routing that feed it.

Evaluation	What it validates	Key metric	Current status
Judge validation	Are the labels consistent enough to train on?	Judge-human Cohen's κ	**Operational**: $\kappa = 0.66$ on the shared benchmark
Value modeling	Can the critic recover meaningful ordinal alignment signal?	QWK, Recall_-1, Minority Recall	**In progress**: current frontier QWK = 0.362 vs 0.40 target
Drift detection	Can weekly routing identify crash/rut patterns reliably?	Hit rate, precision, recall	**Partial**: runtime path exists, benchmark still pending
Explanation quality	Can the system produce grounded, useful feedback?	User-rated usefulness / accuracy	**Partial**: digest generation exists, user-facing evaluation still pending

Judge Validation with Human Annotation

Judge validation currently includes a shared benchmark in the project. Three annotators independently labeled a shared subset of entries using the custom annotation tool, blind to the judge outputs and with chronology preserved within each persona.

On the shared benchmark:

- **Fleiss' $\kappa = 0.56$** for human-human agreement
- **Average Cohen's $\kappa = 0.66$** for judge-human agreement
- the judge exceeds human-human consistency on **9 of 10 dimensions**

This benchmark is the basis for continuing to use judge supervision in the current POC, but it does not settle the question. Follow-up audits showed that agreement alone is not enough: the hardest dimensions can still be unstable or unreachable from the student's available context. That is why the project now treats supervision quality as an active design problem rather than as a solved preprocessing step.

Value Modelling

Value modelling is evaluated at two levels.

At the **entry level**, the key question is whether the critic can classify individual reflections credibly enough to support downstream logic. QWK is the main ranking metric because it respects the ordinal structure of {-1, 0, +1}, while minority-signal metrics reveal whether the model can recover the rare but important misalignment cases Twinkl exists to surface.

At the **persona level**, the question is whether aggregated entry scores can recover the broader value profile implied by a journal trajectory. That remains an important downstream goal, but the current bottleneck is still entry-level reliability. Until that improves further, persona-level ranking is better treated as a follow-on evaluation than as the main proof point for this milestone.

Drift Detection

Drift detection is currently evaluated as a weekly crash/rut-style routing task over aggregated critic outputs and uncertainty estimates. The implementation is in place, but the benchmark is not. The team still needs synthetic crisis-injection timelines, calibrated thresholds, and measured hit rate/precision/recall before this layer can be treated as validated.

Explainable Feedback via Coach

Explanation quality is evaluated by asking whether weekly feedback is grounded, clear, and useful. The project already supports structured digest generation and first-pass automated checks, but user-facing quality targets remain ahead of the current evidence. For this milestone, the explanation path is implemented enough to inspect and improve internally, but it is not yet validated with user ratings.

6. Project Progress and Results

This milestone adds a complete offline supervision pipeline, a validated human benchmark, an experiment archive, and an internal runtime path that can turn a saved checkpoint into weekly review artifacts.

What is runnable today

- Generate synthetic personas and longitudinal journal data
- Produce judge labels with rationales and consolidate them into a training set
- Run human annotation and agreement analysis on a shared benchmark
- Train and compare critic families on fixed holdouts
- Load a saved checkpoint, score a persona timeline, aggregate weekly signals, route crash/rut modes, and generate weekly digest artifacts for internal review

Judge Validation

Before relying on LLM-generated supervision at scale, the team expanded the human benchmark from a small initial sample to **115 shared entries across 19 personas**.

Metric	Value	Interpretation
Shared annotation subset	115 entries across 19 personas	Final like-for-like benchmark
Human-Human Agreement (Fleiss' κ)	0.56	Moderate
Judge-Human Agreement (Avg Cohen's κ)	0.66	Substantial
Dimensions where Judge > Human-Human	9 / 10	Current benchmark used for POC supervision

Later audits refined that conclusion:

Follow-up audit	Scope	Key result	Practical implication
Reachability audit	50 hard cases	Security labels, especially positive ones, were often not reproducible from student-visible context	Part of the bottleneck sits in the target as well as the student
Consensus re-judging	5 judge passes over all 1,651 entries	Judge self-consistency was strong, but the relabeled holdout changed too much for clean replacement of the persisted-label benchmark	Consensus labels are useful diagnostics, not yet a direct frontier replacement

Judge-human agreement varies across dimensions, so supervision quality still needs to be interpreted dimension by dimension.

VIF Experiment Progression

The model search is now large enough to support disciplined comparison rather than isolated intuition. The archive covers **50 run IDs** and **114 persisted configurations**, spanning multiple ordinal losses, calibration strategies, representation choices, and reformulations.

Stage	Main work completed	Main insight
Baseline search	Compared multiple ordinal and long-tail training families	The early bottleneck was neutral-class hedging and weak recovery of misalignment
Evaluation reset	Corrected the validation/test split regime	Earlier leaderboard claims were invalidated; the project reset onto a fairer benchmark
Frontier discovery	Established a BalancedSoftmax family as the corrected-split reference baseline	Early gains came from reducing neutral collapse without abandoning QWK entirely
Systematic follow-up	Tested weighting, encoder changes, two-stage reformulation, and relabeling diagnostics	Several ideas improved specific metrics, but none yet replaced the incumbent cleanly

The frontier is narrower, the remaining questions are better isolated, and the project has evidence for why several alternatives did not become the default.

Current Critic VIF Frontier

The active reference point is the current BalancedSoftmax family under the corrected-split, persisted-label regime.

Metric	Median (3 seeds)
QWK	0.362
Recall_-1	0.313
Minority Recall	0.448
Hedging	62.1%
Calibration	0.713

The current frontier remains below the report's target range for end-user reliability, but the critic is clearly learning signal. The next round of work therefore remains focused on hard dimensions and target quality.

The per-dimension breakdown shows where the critic already captures meaningful signal and where hard dimensions drag the aggregate:

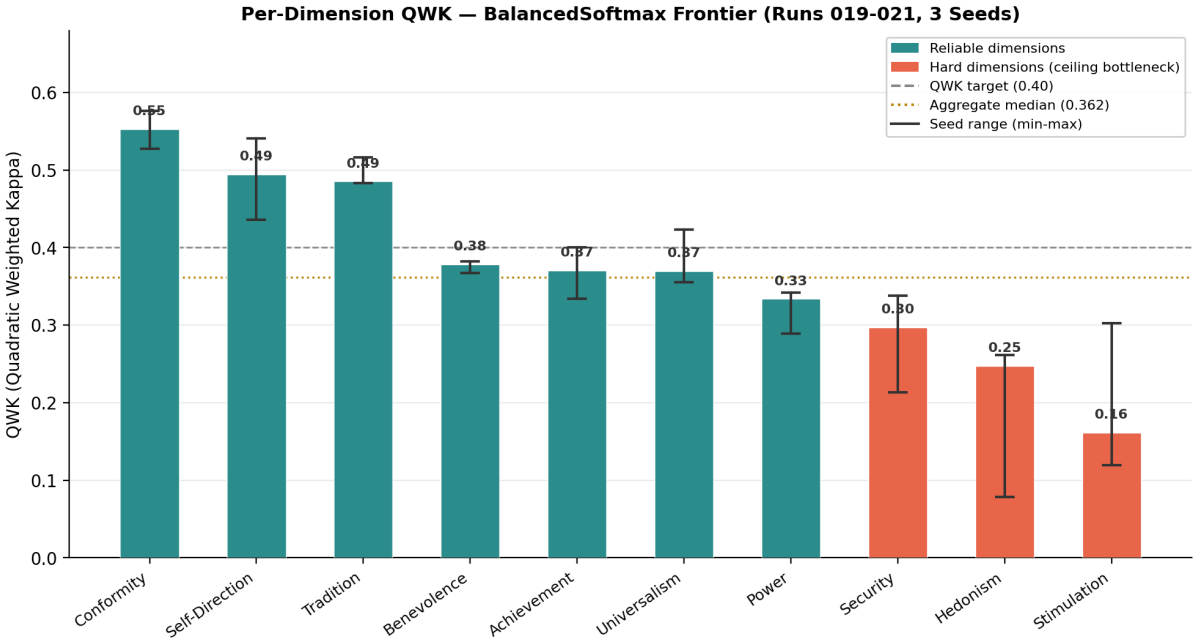


Figure 6: Per-dimension QWK frontier

What Was Tried Beyond the Frontier

The main post-frontier challengers clarified what still does and does not move the system:

Challenger	Best result	Why it did not replace the incumbent
BalancedSoftmax with per-dimension weighting	Best tail-sensitive recovery on misalignment	Improved minority recovery, but overall ranking quality was too unstable
Qwen-based encoder swap	Slightly higher surface QWK and lower hedging	Hard dimensions, especially Hedonism and Power, remained too weak
Two-stage reformulation	Competitive QWK and strong calibration	Became too conservative overall; misalignment recall fell too much
Consensus-label retrain	Better within-regime QWK under relabeled evaluation	Changed the holdout labels, so the result is not a clean like-for-like replacement

One methodological change in this phase is that the project now compares challengers using family-level medians and bootstrap confidence intervals rather than point estimates alone.

Current Findings

The main conclusions at this stage are:

1. **The supervision pipeline is in active use at POC scale**, but hard dimensions still expose target-design weaknesses.

2. **The current critic is the active baseline, but it is not yet a finished model**.
3. **Several nearby alternatives have now been tested and ruled out as default replacements**, which narrows the remaining search space.
4. **The remaining ceiling appears to include both representation and target-quality issues**, with Security as the most visible example of the latter.

The next phase should focus on hard-dimension targets, compact context, and better alignment between training signals and frontier metrics, instead of returning to broad model search.

Annotation Tool

Beyond the core training stack, the project also includes a functioning annotation workflow and an internal review surface that can inspect a saved model checkpoint all the way through to weekly outputs.

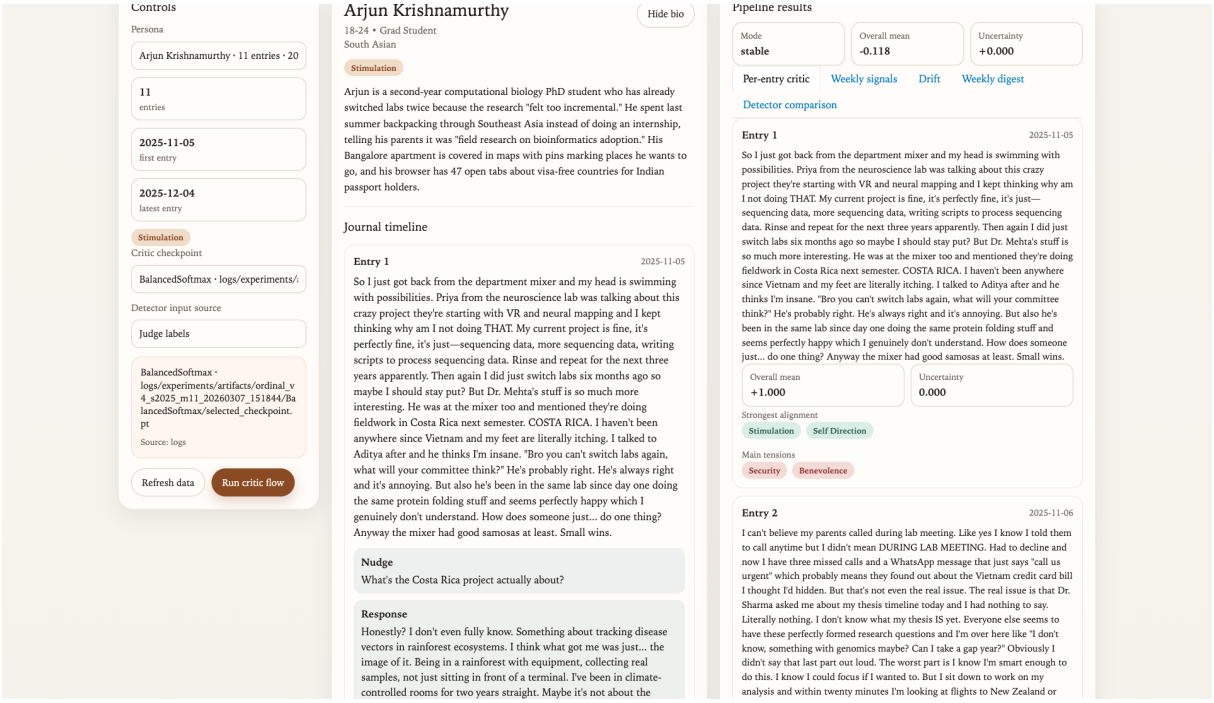


Figure 7: Internal review surface

This makes the pipeline inspectable, debuggable, and demonstrable, even though the current review surface is still internal rather than end-user facing.

The team also built an interactive embedding explorer using Three.js for qualitative inspection of what the critic has learned across all **1,651** labeled entries. The tool offers 3D visualization with multiple projection modes (PCA and t-SNE), per-dimension coloring, individual point inspection, and persona trajectory tracing through the learned representation over time. It serves as an internal review tool for auditing model behaviour beyond aggregate metrics.

Phase 1 (April 2026 Scope-Locking Wave)

As of **6 April 2026**, the project has moved from broad exploration into a scope-locking phase. The central question is now narrower: is the remaining VIF ceiling driven mainly by unreachable hard-dimension targets, by missing short-horizon context, or by limits in the frozen representation?

Workstream	Status	Purpose	Decision it informs
Hard-dimension reachability audit	Completed	Test whether the hardest labels are actually learnable from student-visible context	Whether the current frontier is imitating an inaccessible target
Consensus re-judging diagnostic	Completed	Measure repeated-call stability across the full corpus	Whether label instability is the dominant remaining bottleneck
Matched counterfactual hard-set	Open	Build sharper boundary cases for Hedonism, Security, and Stimulation	Whether the critic still fails on near-counterfactual distinctions after target refinement
Compact history / context prototype	Open	Reintroduce short-horizon context without recreating severe overfitting	Whether the single-entry state is now an artificial ceiling
Training-signal divergence analysis	Open	Test whether current selection signals promote the wrong checkpoints	Whether later training changes should focus on model choice or selection policy

A key completed result is the reachability audit. It showed that aggregate judge-human agreement was not enough to guarantee that the hardest stored labels are clean student targets. The current frontier therefore remains the active baseline, but it is partly a pre-target-redesign reference point rather than the final capstone endpoint.

Integration Milestone

In parallel, the team has outlined a plausible demonstration path. The core path remains:

onboarding -> critic -> weekly digest -> coach

The practical constraint is that the current milestone only supports the middle of that path directly. A messaging-native wrapper remains a plausible Phase 2 delivery option, but it should be treated as a conditional packaging decision rather than as part of the current proven core.

Integration component	Current status	Remaining gap
Onboarding / BWS flow	Documented	Not yet live and not yet wired into active runtime
Critic runtime path	Implemented experimentally	Hard-dimension target redesign still needed before stronger claims
Weekly digest generation	Initial implementation complete	Upstream calibration and evaluation still in progress
Drift-aware routing	Partial	Weekly benchmark calibration remains incomplete
Coach explanation path	Partial	Stronger explanation-to-signal linkage and user-facing evaluation still needed
Confidence-gated triggering	Partial concept plus supporting pieces	Policy layer is not yet finalized as a product behaviour
Messaging-native wrapper	Conditional future direction	No implementation yet; only worth pursuing once core outputs are stable enough to show users

The project can now show a coherent path from declared priorities to scored evidence to weekly reflective output, even though the full end-user journey is not yet complete.

Phase 2 (June-September 2026)

Phase 2 should be treated as a conditional deepening phase, not an automatic continuation of broad model search. The emphasis should depend on what the current scope-locking work proves.

Phase 2 track	Planned work	Why it belongs in Phase 2	Dependency / Gate
Representation follow-up	Add a gated parameter-efficient adaptation path if the frozen encoder is still the main bottleneck	Keeps the model search focused on one remaining technical question	Only after target cleanup and compact-context testing
Drift calibration	Generate synthetic crisis-injection timelines and report hit rate / precision / recall	Converts the current runtime bridge into a properly evaluated weekly routing layer	Requires a stronger upstream critic and a benchmark pass
Narrow messaging-native pilot	Prototype chat-based journaling, cadence checks, and digest delivery	Offers a clear demonstration path without building a separate full application	Only if the core critic and digest outputs are stable enough to surface
External user pilot	Run a small structured validation study with real users to test whether the current alignment and digest outputs transfer beyond synthetic personas and internal review	Converts the largest remaining product risk, external validity, into a measured outcome	Requires stable core outputs and one narrow end-to-end path that is safe to demonstrate
Evolution decision	Decide whether evolution gating remains experimental or enters active scope	Prevents overclaiming long-horizon behaviour change	Depends on the calibrated weekly-routing story
Full end-to-end demonstration	Connect onboarding, scoring, weekly routing, digest generation, and coach output into one clear flow	Moves the project from component validation to system demonstration	Depends on the final scope choice and Phase 1 outcomes

Phase 2 output should include a final system path and a final capstone claim about what Twinkl can and cannot yet do based on the completed work.

Delivery Surfaces and Cost Sketch

The critic model contains approximately **23,000 parameters** and runs **50 MC Dropout forward passes** per inference. At this scale, model footprint and entry-level scoring cost are not the main commercialization constraint. The more significant recurring cost sits in weekly narrative generation rather than in alignment inference itself.

That makes the next packaging question less about raw model deployability and more about delivery surface. Two Phase 2 paths remain credible:

Delivery surface	Why it fits Twinkl	Main trade-off	Current status
Standalone mobile app	Provides a fully owned surface for onboarding, journaling, nudges, and weekly coaching	Highest product and UX implementation burden	Plausible Phase 2 path; not yet built
OpenClaw-based conversational surface	Preserves the same Twinkl core while delivering journaling and weekly review through existing messaging channels, with a natural place for scheduled check-ins and orchestration	Adds a dependency on an external orchestration layer and still requires a scoped wrapper around the current core	Researched integration option; not yet implemented

The project does not need to commit to either path yet. The sponsor-relevant point is that the alignment engine is lightweight enough to support more than one credible delivery surface, while the real Phase 2 decision is which user experience should carry the proven core into a demonstrable end-to-end product.

Deferred and Out-of-Scope

Several ideas remain attractive, but they are not the right focus for the current capstone milestone. The guiding rule is to avoid letting downstream product richness dilute validation of the core judge -> critic -> coach loop.

Deferred area	Current status	Why deferred
Full multi-channel sensing and long-running observability	Conceptual future work	Adds security, privacy, token-cost, and persistence complexity beyond a scoped POC
Habit recommendation system	Parked	Introduces a second intervention problem and requires longitudinal user evidence
Goal-aligned inspiration feed	Not started	Requires external search integration and separate evaluation that do not strengthen the current milestone claim
Multimodal fusion	Future work	Adds major sensing and dataset complexity beyond the text-first capstone scope
Offline RL for nudge or intervention policies	Later-stage idea	Requires stable rewards and longer user trajectories than the project currently has
Adaptive onboarding and dynamic profile refinement	Conceptually important	Depends on validated value-evolution logic and real-user calibration

The project currently prioritizes depth over breadth.

7. Challenges and Open Questions

The project's remaining challenges are concentrated in a small number of hard dimensions and in the quality of their supervision targets. Separately, the biggest product risk is still external validity: most of the current evidence comes from synthetic personas and internal review rather than from real users. Closing that gap through a small external pilot is an explicit Phase 2 deliverable.

Hard Dimensions Still Set the Ceiling

The current frontier is a BalancedSoftmax-based student. Aggregate performance still remains capped by three difficult dimensions: **Security**, **Hedonism**, and **Stimulation**.

These dimensions do not fail for the same reason:

- **Security** appears to be a teacher-student reachability problem. Many labels are not reproducible from student-visible context alone.
- **Hedonism** remains primarily a semantic polarity problem. The model still confuses healthy rest and boundary-setting with avoidance or guilt.
- **Stimulation** is both semantically difficult and statistically weak under the current evaluation regime.

The project has moved past asking only which loss function works best. The more important question now is which parts of the current supervision signal are genuinely learnable from the available inputs.

Label Stability Improved, But Did Not Fully Solve the Problem

Recent work showed clearly that the hard-dimension ceiling is partly a target-quality problem, not only a modelling problem. Consensus re-judging was useful because it proved that the judge can be internally stable on the hard dimensions, but it also showed that greater stability does not automatically produce a better training target.

This changes how the team interprets relabeling results. The team is no longer assuming that more relabeling is automatically better. The current challenge is to improve target quality without erasing the rare, disagreement-heavy edge cases that matter most operationally.

Statistical Power Remains a Real Constraint

The corrected holdout addresses problems in the earlier regime, but it is still small enough that moderate model differences can disappear under proper uncertainty analysis. This means the project must be careful not to overclaim incremental gains, especially on dimensions that remain close to chance-level behaviour.

That constraint also explains why the project now relies on family medians, bootstrap intervals, and promotion criteria.

What Has Been Ruled Out

This phase has already ruled out several simpler explanations. The remaining bottleneck cannot be explained by loss design, generic model capacity, class imbalance, or task formulation alone.

Those negative results narrow the remaining search space and make the next milestone more focused.

Open Questions for the Next Phase

The most important remaining questions are:

1. **What is the right student-reachable supervision target for the hardest dimensions, especially Security?**
2. **Can a compact history representation recover missing signal without making the critic impractical?**
3. **Will a matched hard-set provide a cleaner test of the critic's semantic boundaries than further broad data generation?**
4. **Are current selection signals still misaligned with the frontier metrics the project actually cares about?**
5. **How much of the current synthetic-persona performance transfers to real journaling behaviour once the project begins external validation?**

8. Conclusion

By this milestone, the project has delivered a functioning offline supervision pipeline, a human benchmark, a disciplined critic-training program, and an internal runtime path that produces weekly review artifacts from saved checkpoints. The engineering foundation — 53 test modules, nine ordinal loss variants, six drift detectors, and a four-mode coaching engine — is deeper than a typical POC and designed to support continued iteration.

The honest gaps remain: the critic has not yet reached the QWK target on the hardest dimensions, the weekly drift benchmark is not yet calibrated, and no real user has seen the output. These are the right problems to have at this stage — they are measurable, scoped, and addressed by the Phase 2 plan.

Three concrete steps define the path forward:

1. **Improve hard-dimension supervision targets** so the critic trains against labels that are actually reachable from student-visible context.
2. **Calibrate the weekly routing layer** against synthetic crisis-injection timelines with measured hit rate, precision, and recall.
3. **Run a narrow external pilot** to test whether alignment signals and weekly digests transfer from synthetic personas to real journaling behaviour.

Risk mitigation is built into the timeline. If hard-dimension targets do not improve sufficiently, the project will narrow the active value set to the dimensions where the critic already performs credibly — preserving a demonstrable end-to-end path while being transparent about scope. If the external pilot reveals systematic gaps, the team will treat those findings as the capstone's concluding contribution rather than as a failure.

The team is committed to delivering a working end-to-end demonstration by the final milestone: from declared priorities to scored evidence to a weekly reflective output that a real user can evaluate.

Appendix: Program Submodule Mapping

The following table maps the project's main technical contributions to the practice module's four capability areas.

Submodule	Twinkl Mapping
Pattern Recognition	Ordinal value classification, uncertainty estimation, and sequence-aware weekly aggregation
Intelligent Sensing	Text-derived signals from journal content, temporal patterns, and declared priority profiles
Intelligent Reasoning	Weekly routing logic, evidence-grounded explanation, and profile-conditioned interpretation
Architecting AI Systems	End-to-end orchestration from offline supervision to runtime scoring, weekly review, and future user-facing delivery